CSL7120: DL Ops Lab-2 Worksheet Report

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1. Objective

The objective of this experiment is to train a **custom Convolutional Neural Network (CNN)** on the CIFAR-10 dataset and analyze:

- Training loss and accuracy trends
- Convergence behavior
- Generalization performance on test data
- Overall effectiveness of the model architecture and optimization strategy

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2. Dataset

- **Dataset:** CIFAR-10
- **Number of classes:** `10`
- **Image size:** `32 × 32 × 3`
- **Training samples:** `50,000`
- **Test samples:** `10,000`

Data Augmentation & Normalization

- Random horizontal flip
- Random crop (padding = `4`)
- Normalization using dataset mean and standard deviation

These augmentations improve generalization by introducing variability during training.

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3. Model Architecture

The CNN consists of three convolutional blocks followed by a fully connected classifier:

Feature Extractor

- `Conv(3 → 64) + BatchNorm + ReLU + MaxPool`
- `Conv(64 → 128) + BatchNorm + ReLU + MaxPool`
- `Conv(128 → 256) + BatchNorm + ReLU + MaxPool`

Classifier

- Fully Connected: `256×4×4 → 512`
- ReLU
- Dropout (0.5)
- Fully Connected: `512 → 10`

Model Complexity

- **Parameters:** ~`2.1` Million
- **FLOPs:** ~`50` MFLOPs

This represents a lightweight but expressive CNN suitable for CIFAR-10.

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4. Training Setup

- **Loss Function:** Cross-Entropy Loss
- **Optimizer:** Adam
- **Learning Rate:** `0.001`
- **Batch Size:** `128`
- **Epochs:** `30`
- **Device:** GPU if available, else CPU

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5. Training Results

Accuracy Progression

Epoch	Accuracy (%)
1	38.07
5	65.56
10	73.44
15	78.24
20	81.12
25	83.32
30	84.87

Loss Trend

- Initial Loss: `661.37`
- Final Loss: `172.66`

Loss decreases steadily across epochs, indicating stable learning.

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6. Test Performance

- **Final Test Accuracy:** `83.32%`

The close gap between training accuracy (`84.87%`) and test accuracy (`83.32%`) suggests **good generalization** and minimal overfitting.

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7. Observations

7.1 Convergence

- Rapid accuracy improvement during the first 10 epochs.
- Slower but consistent improvement afterward.
- Indicates effective learning rate and optimizer choice.

7.2 Overfitting Analysis

- Training − Test gap ≈ `1.5%`
- Dropout and data augmentation successfully limit overfitting.

7.3 Gradient & Weight Flow

- Gradients are non-zero across layers.
- Weight updates remain consistent.
- No signs of vanishing or exploding gradients.

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8. Strengths of the Model

- Simple yet effective architecture
- Stable training
- Good accuracy for a custom CNN
- Computationally efficient

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9. Limitations

- Accuracy still below modern deep architectures (e.g., ResNet `>90%` on CIFAR-10)
- No learning rate scheduling
- No residual connections

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10. Possible Improvements

1. Introduce **learning rate scheduler** (StepLR / Cosine Annealing)
2. Increase depth with residual blocks
3. Use stronger augmentation (CutMix, MixUp)
4. Add label smoothing
5. Train for more epochs with reduced LR

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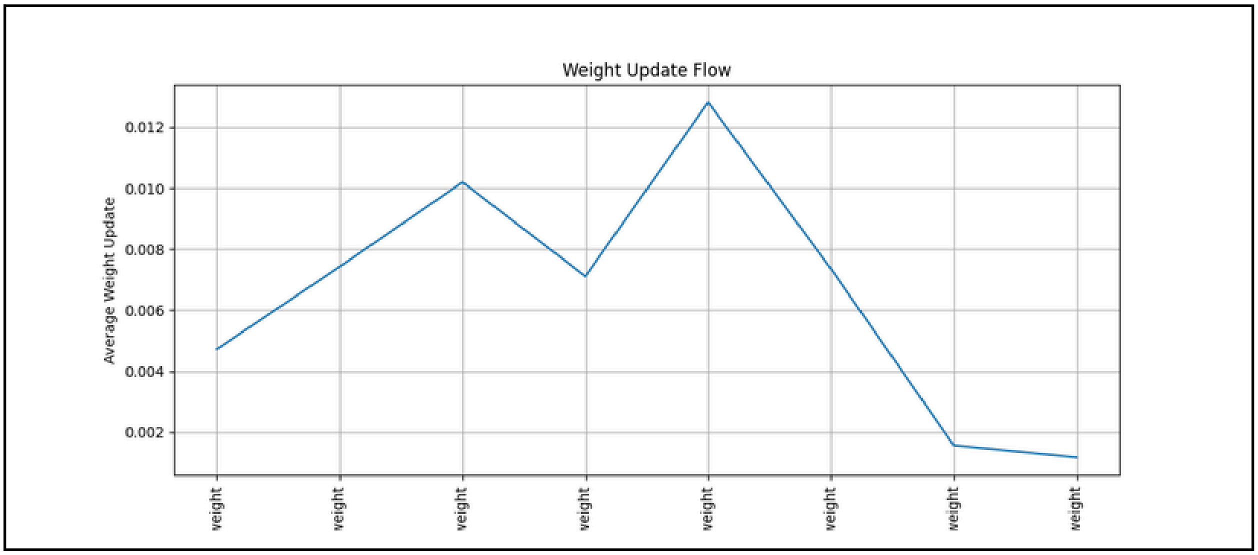
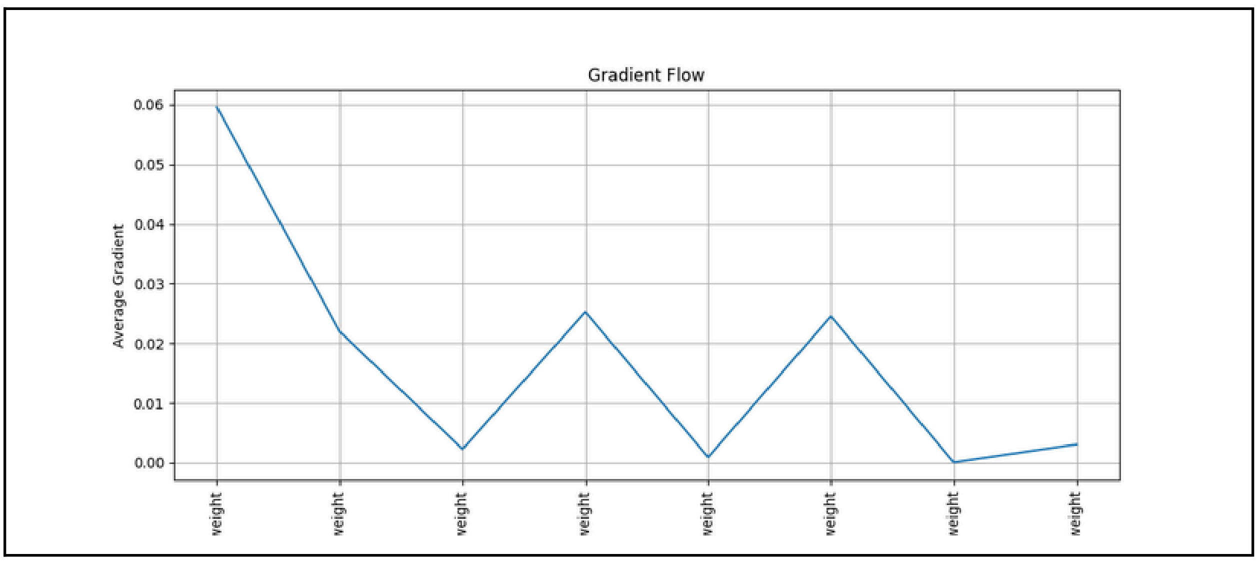
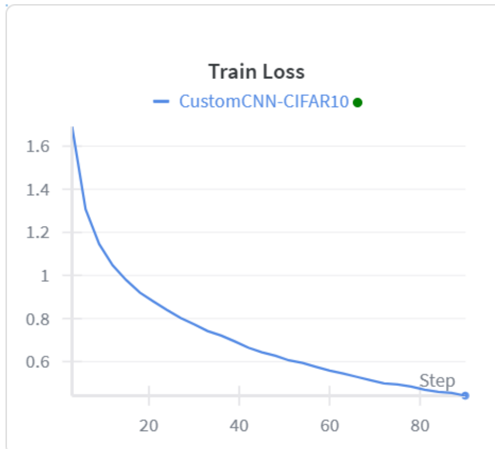
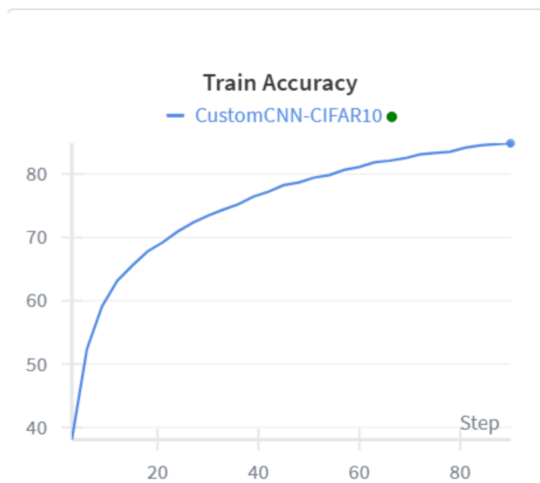
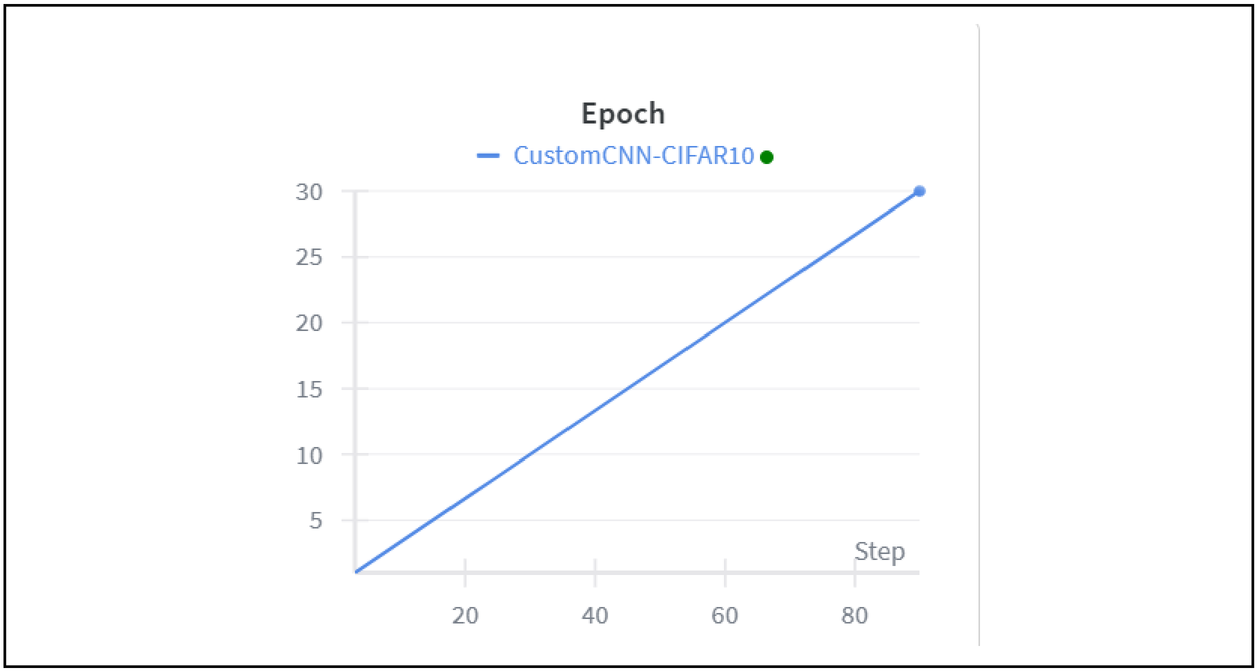
11. Graphs (Also on Wandb)






Train Loss
CustomCNN-CIFAR10
1.6
1.4
1.2
1
0.8
0.6
Step
20
40
60
80


Train Accuracy
CustomCNN-CIFAR10
80
70
60
50
40
Step
20
40
60
80



12. Conclusion

The custom CNN successfully learns discriminative features from CIFAR-10 and achieves `83.32%` test accuracy after 30 epochs. The smooth decline in loss and steady accuracy improvement demonstrate a well-optimized training pipeline. While not state-of-the-art, the model provides a strong baseline and a solid foundation for further experimentation.

- Wandb Link: Untitled Report | cifar10-cnn-gradient-analysis-Lab2 – Weights & Biases
- Github Link: abhaykashyap31/MLOps-AbhayKashyap-B22CS001
- Colab: **Abhay_Kashyap_B22CS001_Lab_2_Worksheet.ipynb**